Nanxiang Jiang

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Education

Beihang University (BUAA), Beijing, China *2023 – Present*
B.E. in Computer Science (Expected Jun 2027), Top Talents Program in Computer Science
GPA: 3.86 / 4.0 (Rank: 2 / 20); Weighted Average: 92.9 / 100 (Rank: 1 / 20)

Research Interest

My primary research goal is to **build intelligent agents capable of understanding, reconstructing, and simulating the physical world**. I am particularly interested in agentic approaches to physical world reasoning, inverse graphics and dynamics, multimodal generative models, and action-conditioned world modeling.

Publications and Preprints

(* Equal Contribution; † Equal Advising)

[1] **Inverse Garment Design with Visual Agents**
Under Review [Paper] [Project] [Slides]
Nanxiang Jiang, Yunuo Chen, Zeng Tao, Zizhang Li, Qi Guo, Jiajun Wu, Yin Yang, Chenfanfu Jiang

[2] **Localizing Concepts in Visual Autoregressive Models**
NeurIPS 2026 [Paper]
Nanxiang Jiang, Yang Liu, Yuanhao Wang, Min Xu

[3] **Z-Erase: Enabling Concept Erasure in Single-Stream Diffusion Transformers**
ICML 2026 [Paper] [Project] [Code]
Nanxiang Jiang, Zhaoxin Fan, Baisen Wang, Daiheng Gao, Junhang Cheng, Jifeng Guo, Yalan Qin, Yeying Jin, Hongwei Zheng, Faguo Wu, Wenjun Wu

[4] **Erased, But Not Forgotten: Erased Rectified Flow Transformers Still Remain Unsafe Under Concept Attack**
CVPR Findings 2026 [Paper] [Code]
Nanxiang Jiang, Zhaoxin Fan, Enhan Kang, Daiheng Gao, Yun Zhou, Yanxia Chang, Zheng Zhu, Yeying Jin, Wenjun Wu

[5] **RoboPARA: Dual-Arm Robot Planning with Parallel Allocation and Recomposition Across Tasks**
ICLR 2026 [Paper] [Code]
Shiying Duan, Pei Ren*, Nanxiang Jiang, Zhengping Che, Jian Tang, Yifan Sun, Zhaoxin Fan, Wenjun Wu*

[6] **CangjieBench: Benchmarking LLMs on a Low-Resource General-Purpose Programming Language**
ESEM 2026 [Paper] [Code]
Junhang Cheng, Fang Liu, Jia Li, Chengru Wu, Nanxiang Jiang†, Li Zhang†

[7] **EraseAnything++: Enabling Concept Erasure in Rectified Flow Transformers Leveraging Multi-Object Optimization**
Under Review [Paper] [Code]
Zhaoxin Fan, Nanxiang Jiang*, Daiheng Gao, Shiji Zhou, Wenjun Wu*

[8] **Revoking Amnesia: RL-based Trajectory Optimization to Resurrect Erased Concepts in Diffusion Models**
Under Review [Paper]
Daiheng Gao, Nanxiang Jiang*, Andi Zhang, Shilin Lu, Yufei Tang, Wenbo Zhou, Weiming Zhang, Zhaoxin Fan*

[9] **Understanding Variable Length Arrays in the C Language**
IEEE Software [Paper]
Zhiwei Sun, Wenge Rong, Nanxiang Jiang, Yuxi Bai, Jing Li, Zhang Xiong

[10] **Global Nonlinear Simulation of Frequency Sweeping of Hot Electron Interchange Modes in a Dipole-Confined Plasma**
Under Review [Paper]
Zikang Yang, Nanxiang Jiang*, Lipeng Wang, Changzhi Jiang, Tianchun Zhou, Bo Li, M. E. Mauel*

Selected Awards

SenseTime Scholarship *(Top 30 nationwide)* *2026*
National Second Prize, Ministry of Education Question & Conjecture Program *(Top 15 nationwide)* *2026*
National Scholarship *(Highest honor for undergraduates in China)* *2024*

Research Experience

University of California, Los Angeles *May 2026 – Sep. 2026*
Research Intern, advised by Prof. Chenfanfu Jiang

- Led DressAgent, a multi-agent system recovering simulation-ready layered 3D outfits from one image via closed-loop planning and refinement, with the first 15K+ multi-layer garment dataset (Under Review, first author) [1].

Carnegie Mellon University *Feb. 2026 – May 2026*
Research Intern, advised by Prof. Min Xu

- Led LoCo, the first framework to analyze where and when concepts emerge (layer, scale, position) in image and video AR models, enabling SOTA low-cost erasure, personalization, and injection (NeurIPS 2026, first author) [2].

Beihang University *Jul. 2024 – Jan. 2026*
Research Assistant at CoLab, advised by Prof. Si Liu and Prof. Zhaoxin Fan

- Led Z-Erase, enabling concept erasure in single-stream image generation DiTs (e.g., Z-Image) via stream-decoupled updates and a Lagrangian erasure–utility trade-off with convergence analysis (ICML 2026, first author) [3].
- Led ReFlux, a reverse-attention attack method showing that erasure in Flux hinges on attention localization, providing a benchmark for erasure robustness (CVPR Findings 2026, first author) [4].
- Designed the constrained multi-objective optimization behind EraseAnything++, resolving erasure–utility gradient conflicts via implicit gradient surgery on SD3, Flux, and OpenSora (Under Review, co-first author) [7].
- Devised the GRPO adaptation behind RevAm, casting denoising as policy optimization with trajectory-level rewards to probe erasure durability in image generation, at ∼10× less compute (Under Review, co-first author) [8].

Beihang University *Jan. 2025 – Sep. 2025*
Research Assistant at School of Physics, advised by Prof. Bo Li

- Implemented the core MATLAB simulations for a physics team modeling how hot electron interchange instability in dipole-confined plasma saturates and sweeps up in frequency (Under Review, co-first author) [10].

Industry Experience

X Humanoid (Beijing Humanoid Robot Innovation Center) *Apr. 2025 – Sep. 2025*
Research Intern, advised by Pei Ren

- Designed the DAG traversal algorithm behind RoboPARA, recomposing task dependency graphs to maximize parallel execution in LLM-driven dual-arm planning, and led the main experiments (ICLR 2026) [5].

Academic Services

Reviewer CVPR, ICML, ICME *2025 – Present*
Teaching Assistant C Language Programming (co-designed its VLA module, IEEE Software [9]) *Fall 2024*

Skills

Languages Chinese (Native), English (TOEFL iBT: 110, 5.5/6)
Programming Python, PyTorch, Java, C/C++, MATLAB, NVIDIA Warp, Blender, Diffusers